

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

January 2023

20EC501 – ANALOG COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
2) Assume missing data suitably.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Let X and Y be defined by $X = \cos \theta$ and $Y = \sin \theta$, where θ is a random variable uniformly distributed over $[0, 2\pi]$. (i) Determine the mean values of X and Y (ii) Show that X and Y are uncorrelated.	8	L*3	1	1
b) Define the following: (i) mean (ii) correlation (iii) covariance	6	L2	1	1
c) The probability density function of random variable x is given as $f_x(x) = k \cdot e^{-2x} u(x)$. A random variable y is defined as $y = 5x + 6$. Determine the probability density function of the random variable y.	6	L3	1	1
2. a) Two random processes X(t) and Y(t) given by $X(t) = A \cos(\omega t + \theta)$ and $Y(t) = A \sin(\omega t + \theta)$, where A and ω are constants and the random variable θ is uniformly distributed over the interval $(0, 2\pi)$. Determine the cross correlation function of processes X(t) and Y(t) and also prove that $R_{xy}(-\tau) = R_{yx}(\tau)$.	8	L3	1	1
b) Let X be a random variable which is uniformly distributed in the interval $(-\pi, \pi)$. If $Y = \cos X$, Determine: (i) mean value of Y (ii) mean squared value of Y (iii) variance of Y	6	L3	1	1
c) The pdf of a random variable X is given by $f_x(x) = k$, $a \leq x \leq b$, and zero otherwise, where k is a constant. Determine (i) the value of k (ii) let $a = -1$ and $b = 2$. Calculate the $P(X \leq c)$ for $c = \frac{1}{2}$.	6	L3	1	1

Unit – II

3. a) Derive the expression for the single sideband modulated signal where only upper sideband is transmitted. Also draw the block diagram for the modulator.	8	L2	2	1
b) Explain the generation of AM using square law modulator with relevant equations.	6	L2	2	1
c) Consider a message signal $m(t) = 30 \cos 2\pi t$ volts and the carrier wave $c(t) = 50 \cos 100\pi t$ volts. (i) obtain an expression for the resulting amplitude modulated wave for 65% modulation (ii) find the power dissipated across a load of 600Ω due to this amplitude modulated wave	6	L3	2	1
4. a) An amplitude modulated waveform has the form: $x(t) = 10 \cos 2\pi 10^6 t + 5 \cos 2\pi 10^6 t \cdot \cos 2\pi 10^3 t + 2 \cos 2\pi 10^6 t \cdot \cos 4\pi 10^3 t$ volts. (i) find the various spectral components present (ii) Sketch the amplitude spectra of x(t).	8	L3	2	1
b) Explain the generation of DSB-SC wave using ring modulator with neat diagrams and waveforms.	6	L2	2	1
c) Find the Hilbert transform of : (i) $x(t) = e^{-j2\pi ft}$ (ii) $x(t) = \delta(t)$ (iii) $x(t) = \cos 2\pi ft + \sin 2\pi ft$	6	L3	2	1

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

January 2023

20EC501 – ANALOG COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
2) Assume missing data suitably.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Let X and Y be defined by $X = \cos \theta$ and $Y = \sin \theta$, where θ is a random variable uniformly distributed over $[0, 2\pi]$. (i) Determine the mean values of X and Y (ii) Show that X and Y are uncorrelated.	8	L*3	1	1
b) Define the following: (i) mean (ii) correlation (iii) covariance	6	L2	1	1
c) The probability density function of random variable x is given as $f_x(x) = k \cdot e^{-2x} u(x)$. A random variable y is defined as $y = 5x + 6$. Determine the probability density function of the random variable y.	6	L3	1	1
2. a) Two random processes X(t) and Y(t) given by $X(t) = A \cos(\omega t + \theta)$ and $Y(t) = A \sin(\omega t + \theta)$, where A and ω are constants and the random variable θ is uniformly distributed over the interval $(0, 2\pi)$. Determine the cross correlation function of processes X(t) and Y(t) and also prove that $R_{xy}(-\tau) = R_{yx}(\tau)$.	8	L3	1	1
b) Let X be a random variable which is uniformly distributed in the interval $(-\pi, \pi)$. If $Y = \cos X$, Determine: (i) mean value of Y (ii) mean squared value of Y (iii) variance of Y	6	L3	1	1
c) The pdf of a random variable X is given by $f_x(x) = k$, $a \leq x \leq b$, and zero otherwise, where k is a constant. Determine (i) the value of k (ii) let $a = -1$ and $b = 2$. Calculate the $P(X \leq c)$ for $c = \frac{1}{2}$.	6	L3	1	1

Unit – II

3. a) Derive the expression for the single sideband modulated signal where only upper sideband is transmitted. Also draw the block diagram for the modulator.	8	L2	2	1
b) Explain the generation of AM using square law modulator with relevant equations.	6	L2	2	1
c) Consider a message signal $m(t) = 30 \cos 2\pi t$ volts and the carrier wave $c(t) = 50 \cos 100\pi t$ volts. (i) obtain an expression for the resulting amplitude modulated wave for 65% modulation (ii) find the power dissipated across a load of 600Ω due to this amplitude modulated wave	6	L3	2	1
4. a) An amplitude modulated waveform has the form: $x(t) = 10 \cos 2\pi 10^6 t + 5 \cos 2\pi 10^6 t \cdot \cos 2\pi 10^3 t + 2 \cos 2\pi 10^6 t \cdot \cos 4\pi 10^3 t$ volts. (i) find the various spectral components present (ii) Sketch the amplitude spectra of x(t).	8	L3	2	1
b) Explain the generation of DSB-SC wave using ring modulator with neat diagrams and waveforms.	6	L2	2	1
c) Find the Hilbert transform of : (i) $x(t) = e^{-j2\pi ft}$ (ii) $x(t) = \delta(t)$ (iii) $x(t) = \cos 2\pi ft + \sin 2\pi ft$	6	L3	2	1

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (E&C) (Credit System) Degree Examinations**

January 2023

20EC501 – ANALOG COMMUNICATION

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) Assume missing data suitably.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) Let X and Y be defined by $X = \cos \theta$ and $Y = \sin \theta$, where θ is a random variable uniformly distributed over $[0, 2\pi]$.
(i) Determine the mean values of X and Y
(ii) Show that X and Y are uncorrelated. | 8 | L*3 | 1 | 1 |
| b) Define the following: (i) mean (ii) correlation (iii) covariance | 6 | L2 | 1 | 1 |
| c) The probability density function of random variable x is given as $f_x(x) = k.e^{-2x}u(x)$. A random variable y is defined as $y = 5x + 6$. Determine the probability density function of the random variable y. | 6 | L3 | 1 | 1 |
| 2. a) Two random processes X(t) and Y(t) given by $X(t) = A \cos(\omega t + \theta)$ and $Y(t) = A \sin(\omega t + \theta)$, where A and ω are constants and the random variable θ is uniformly distributed over the interval $(0, 2\pi)$. Determine the cross correlation function of processes X(t) and Y(t) and also prove that $R_{xy}(-\tau) = R_{yx}(\tau)$. | 8 | L3 | 1 | 1 |
| b) Let X be a random variable which is uniformly distributed in the interval $(-\pi, \pi)$. If $Y = \cos X$, Determine: (i) mean value of Y (ii) mean squared value of Y (iii) variance of Y | 6 | L3 | 1 | 1 |
| c) The pdf of a random variable X is given by $f_x(x) = k$, $a \leq x \leq b$, and zero otherwise, where k is a constant. Determine (i) the value of k (ii) let $a = -1$ and $b = 2$. Calculate the $P(X \leq c)$ for $c = \frac{1}{2}$. | 6 | L3 | 1 | 1 |

Unit – II

- | | | | | |
|---|---|----|---|---|
| 3. a) Derive the expression for the single sideband modulated signal where only upper sideband is transmitted. Also draw the block diagram for the modulator. | 8 | L2 | 2 | 1 |
| b) Explain the generation of AM using square law modulator with relevant equations. | 6 | L2 | 2 | 1 |
| c) Consider a message signal $m(t) = 30 \cos 2\pi t$ volts and the carrier wave $c(t) = 50 \cos 100\pi t$ volts.
(i) obtain an expression for the resulting amplitude modulated wave for 65% modulation
(ii) find the power dissipated across a load of 600Ω due to this amplitude modulated wave | 6 | L3 | 2 | 1 |
| 4. a) An amplitude modulated waveform has the form:
$x(t) = 10 \cos 2\pi 10^6 t + 5 \cos 2\pi 10^6 t \cdot \cos 2\pi 10^3 t + 2 \cos 2\pi 10^6 t \cdot \cos 4\pi 10^3 t$ volts. (i) find the various spectral components present
(ii) Sketch the amplitude spectra of x(t). | 8 | L3 | 2 | 1 |
| b) Explain the generation of DSB-SC wave using ring modulator with neat diagrams and waveforms. | 6 | L2 | 2 | 1 |
| c) Find the Hilbert transform of :
(i) $x(t) = e^{j2\pi ft}$
(ii) $x(t) = \delta(t)$
(iii) $x(t) = \cos 2\pi ft + \sin 2\pi ft$ | 6 | L3 | 2 | 1 |

- b) Two two-port devices are connected in cascade. For the first stage, the noise figure and available power gain are 5 dB and 12 dB respectively. For the second stage, the noise figure and available power gain are 15 dB and 10 dB respectively. Determine the overall noise figure in dB.
- c) Explain briefly: (i) Thermal noise (ii) Shot noise.

6	L3	4	1
6	L2	4	1

Unit – V

9. a) Discuss the noise in DSB-SC receivers using coherent detection scheme and derive the expression for figure of merit of the above scheme.
- b) The carrier reaching an envelope detector in an AM receiver has an RMS value equal to 1 volt in the absence of modulation. The noise at the input of the envelope detector has a PSD equal to 10^{-3} watts/Hz. The carrier is modulated to a depth of 100% and message bandwidth, $W = 3.2$ kHz. Calculate the $[SNR]_o$.
- c) With neat diagrams explain pre-emphasis and de-emphasis networks.
10. a) Discuss the noise in SSB receiver with a model receiver using coherent detection. Derive the figure of merit of the receiver.
- b) Explain the receiver structure for conventional AM wave and hence derive the expression for figure of merit.

8	L2	5	1
---	----	---	---

6	L3	5	1
---	----	---	---

6	L2	5	1
---	----	---	---

10	L2	5	1
----	----	---	---

10	L2	5	1
----	----	---	---

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
